

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/od120lite, x86-64 Linux, GCC 15

Verdict

The capability does not transfer to ordinary objects under ordinary tracking. Hardware-block separation survives about **2 m** of position noise on a 22-hour arc and about **0.1 m** on a six-hour one; real tracking sits one to two orders of magnitude outside that. The general space situational awareness thesis is closed. One narrower version survives, untested: the objects where this estimator would be most precise are exactly the objects where radiation pressure dominates — which are exactly the objects nobody can currently characterise. That is being settled analytically from data already in hand, and it is the last open technical question.

The degradation result

Arcs that work were degraded along three axes until the block separation failed, measured against a 2.75% threshold — half the gap between adjacent hardware blocks.

arc	fractional error on the estimate, by position noise		
22 h hourly ($n = 22$)	1 m: 1.3%	3 m: 4.2%	10 m: 14%
6 h hourly ($n = 7$)	0.1 m: 2.8%	0.3 m: 7.3%	1 m: 23%

Arc length and noise tolerance trade off roughly tenfold. Beyond the surviving region the estimate stops being a measurement at all — at six hours, hourly, 10 m, one satellite returns +226% and another −86%: the parameter is absorbing whatever the geometry allows.

The structure matters more than the number

Taken one axis at a time the picture was misleadingly good. Six observations spread over 22 hours give 0.35%, as good as 88 of them. A one-hour densely sampled arc holds 2.3%. Neither sparsity nor arc length binds. **Noise binds, and it binds far harder on a short arc** — it is the interaction that kills the capability, and three independent sweeps would have reported a robust one.

That is the most transferable finding in this engagement, and it is the fourth time this codebase's failure mode has been a confident wrong answer rather than an error. The others: a least squares built on identity partials, a header-line count that silently restricted validation to one constellation, an assumed sampling interval, and observations matched to states by index on the assumption of a complete uniform reference. All are now fixed, and that pattern — *fails quietly rather than loudly* — is the single most important thing for anyone inheriting this code to know.

The uncertainty does not support the estimate

Formal parameter sigma from the fit is 0.01–0.035%. Empirical scatter is 0.29% within the largest block and 0–0.5% across dates. The formal figure is optimistic by 10–20×.

The cause is diagnosed: the weight matrix is identity on positions, asserting independent white noise, while the residuals are systematic and correlated along the arc — the signature of unmodelled dynamics. So the formal sigma cannot carry a decision threshold, and the degradation study was correctly measured against empirical deviation instead.

Reporting it anyway was right. The disagreement is itself a measurement of how far the noise

model sits from the truth, which is more useful than a sigma that happened to agree.

What is closed, and what is not

Closed: the general SSA thesis. Characterising ordinary catalogued objects from ordinary tracking is outside the surviving region by one to two orders of magnitude in noise. No amount of engineering on this codebase changes that, because the limit is information in the observations, not software.

Not closed, and worth one more test: the high area-to-mass case. The estimator recovers an effective AC_R/m , and the signal it works from is the radiation-pressure displacement accumulated over the arc. GNSS sits near $0.02 \text{ m}^2/\text{kg}$. High area-to-mass debris runs roughly 1 to 30 — fifty to fifteen hundred times the signal in the same residual. If precision scales inversely with signal, the 2 m threshold becomes a very different number.

The symmetry is the point: the objects this estimator would measure best are the ones where radiation pressure dominates the dynamics, which are the ones that are hardest to characterise by any other means. That is not a coincidence; it is the same physics read from both ends.

Two things cut against it, and both are being tested rather than assumed. The estimator requires AC_R/m to be constant over the arc, and a tumbling or irregular object's effective area is not — although an estimator that *detects* that variation rather than silently averaging it would be a more useful instrument, and the residual may carry the signature. And the degraded-GNSS noise model is white Gaussian on Cartesian position, not angles and range from a station with correlated errors; the faithful version of that test needs observation models, which remain deliberately unbuilt.

Where the asset stands

Revision 1	The ray-traced SRP engine — scarce, and missing its fuel
Revision 2	The validation — a credential making the SRP claim falsifiable
Revision 3	A parameter-estimating orbit determination capability
Revision 4	A robust estimator that may characterise objects from dynamics alone
Revision 5	It does — for large, well-tracked objects
Revision 6	And only for those. The commercial regime is out of reach on current evidence, with one specific exception under test.

Six revisions, each moved by evidence rather than by rearranging the framing. Three of them moved against the thesis. That record is the reason the remaining claims can be put in front of someone.

Valuation

- **As an orbit determination product: approximately £0.** Unchanged throughout, and nothing found since has challenged it.
- **As an SSA characterisation capability: approximately £0 on current evidence,** pending the high area-to-mass test. This was the one thesis that could have carried a business and it is now one test from closed.
- **As a validated body of work: genuinely positive, and the durable result.** A tree that did not build is now under version control with documented provenance, an honest accuracy figure against public reference data, a robust estimator, five silent-wrong-answer defects found and fixed, and a published-quality finding — that spacecraft hardware blocks are recoverable

from dynamics alone. That is a real asset for publication, teaching and credibility, and it is not nothing.

- **Replacement cost remains the wrong frame** at £300k–£600k.

Recommendation

This is a decision point rather than another phase. One technical phase remains and it is scoped to close the last open question; after it, the choice is commercial and yours.

1. **Settle the high area-to-mass question analytically.** *In progress* — extrapolating the measured degradation surface to higher signal, and testing whether a tumbling object breaks the constant-parameter premise. Both from data in hand.
2. **If it clears:** the single remaining investment worth making is observation models and covariance, and it should not start before the IP position is settled. The decisive validation would still need a high area-to-mass ephemeris, which is currently blocked on credentialed data access.
3. **If it does not: stop, and publish.** The hardware-block result is interesting on its own terms and is the natural output. Finishing there is a legitimate outcome, not a failure.
4. **Settle the IP position regardless.** It gates publication as surely as it gates sale, it has been outstanding since the first revision, and it needs no engineering at all.

Basis: measured facts come from the tree at `/home/rog/odl20lite` and from runs executed on 15 September 2026 by the `odl` maintainer session, reviewed and spot-verified here. The competitive and valuation sections are judgement. Revision 6 supersedes revision 5, which reported the characterisation capability without knowing the observational regime it requires.